

## Topic 8: Nuclear and Particle Physics

Mathematical skills that could be developed in this topic include using appropriate units in calculations.

| Students should: |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| 130.             | understand what is meant by <i>nucleon number (mass number)</i> and <i>proton number (atomic number)</i>                                                                                                                                                                                                                                                                                                                                          |
| 131.             | understand how large-angle alpha particle scattering gives evidence for a nuclear model of the atom and how our understanding of atomic structure has changed over time                                                                                                                                                                                                                                                                           |
| 132.             | understand that electrons are released in the process of thermionic emission and how they can be accelerated by electric and magnetic fields                                                                                                                                                                                                                                                                                                      |
| 133.             | understand the role of electric and magnetic fields in particle accelerators (linac and cyclotron) and detectors (general principles of ionisation and deflection only)                                                                                                                                                                                                                                                                           |
| 134.             | be able to derive and use the equation $r = \frac{p}{BQ}$ for a charged particle in a magnetic field                                                                                                                                                                                                                                                                                                                                              |
| 135.             | be able to apply conservation of charge, energy and momentum to interactions between particles and interpret particle tracks                                                                                                                                                                                                                                                                                                                      |
| 136.             | understand why high energies are required to investigate the structure of nucleons                                                                                                                                                                                                                                                                                                                                                                |
| 137.             | be able to use the equation $\Delta E = c^2 \Delta m$ in situations involving the creation and annihilation of matter and antimatter particles                                                                                                                                                                                                                                                                                                    |
| 138.             | be able to use MeV and GeV (energy) and MeV/c <sup>2</sup> , GeV/c <sup>2</sup> (mass) and convert between these and SI units                                                                                                                                                                                                                                                                                                                     |
| 139.             | understand situations in which the relativistic increase in particle lifetime is significant (use of relativistic equations not required)                                                                                                                                                                                                                                                                                                         |
| 140.             | know that in the standard quark-lepton model particles can be classified as: <ul style="list-style-type: none"> <li>• baryons (e.g. neutrons and protons) which are made from three quarks</li> <li>• mesons (e.g. pions) which are made from a quark and an antiquark</li> <li>• leptons (e.g. electrons and neutrinos) which are fundamental particles</li> <li>• photons</li> </ul> and that the symmetry of the model predicted the top quark |
| 141.             | know that every particle has a corresponding antiparticle and be able to use the properties of a particle to deduce the properties of its antiparticle and vice versa                                                                                                                                                                                                                                                                             |
| 142.             | understand how to use laws of conservation of charge, baryon number and lepton number to determine whether a particle interaction is possible                                                                                                                                                                                                                                                                                                     |
| 143.             | be able to write and interpret particle equations given the relevant particle symbols.                                                                                                                                                                                                                                                                                                                                                            |